

State Management in ASP.NET Web

Start with previous lab application.

The screenshot displays the Visual Studio IDE with the 'BookingApp' project open. The main window shows the 'BookStay.aspx.designer.cs' file in Design view. The form includes the following elements:

- What's your name?** (Text input field)
- How many adults?** (Text input field)
- You are interested in...** (Radio button group with options: Studio, Suite, Deluxe, Double, Single)
- Actually interested in ...** (List box with options: Swimming Pool, Gym, Tennis court, Horse Riding)
- We enjoy** (Text input field)
- My secret is** (Text input field)
- Please check following:** (List of error messages: Error message 1, Error message 2)

Red validation messages are displayed on the right side of the form:

- Please enter name**
- It can't be 0Please enter a countRange(1-6)**
- What is your food preference?**
- Secret must an alphanumeric string of length 6**

The 'RangeValidator' control is highlighted, and its properties are shown in the Properties window:

Property	Value
MinimumValue	1
SetFocusOnError	False
SkinID	
TabIndex	0
Text	Range(1-6)
ToolTip	
Type	Integer
ValidateRequestMode	Inherit
ValidationGroup	
ViewStateMode	Inherit

The Solution Explorer on the right shows the project structure:

- BookingApp**
 - Connected Services
 - Properties
 - AssemblyInfo.cs
 - References
 - BookStay.aspx
 - BookStay.aspx.cs
 - BookStay.aspx.designer.cs
 - packages.config
 - Web.config

We did it....

localhost:26124/BookStay.aspx x +

localhost:26124/BookStay.aspx

Apps Combining Structur...

What's your name?

How many adults?

You are interested in...

☒ Studio
☐ Suite
☐ Deluxe
☐ Double
☐ Single

Actually interested in ...

☐ Swimming Pool
☐ Gym
☐ Tennis court
☐ Horse Riding

We enjoy

My secret is

Please enter name

Please enter a count

What is your food preference?

Please check following:

- Name is required.
- Count is required.
- Please select your food preference

Windows Taskbar: Type here to search, Task View, Edge, File Explorer, Chrome, VS Code, Outlook, System Tray (Network, Volume, Bluetooth, Date/Time: 10:46 AM 28-Jul-22, Notifications: 2)

We did it....

localhost:26124/BookStay.aspx x +

localhost:26124/BookStay.aspx

Apps Combining Structur...

Thank you, Apurva for your booking.
Apurva your booking for 2 people in our double room is confirmed.
Apurva your food preference is veg.
Apurva your secret is micros.
Apurva, thank you for choosing following facilities.
Gym
Tennis court
Take care.

What's your name?

How many adults?

☐ Studio
☐ Suite

You are interested in... ☐ Deluxe
☒ Double
☐ Single

☐ Swimming Pool

Actually interested in ... ☒ Gym
☒ Tennis court
☐ Horse Riding

We enjoy

My secret is

Type here to search

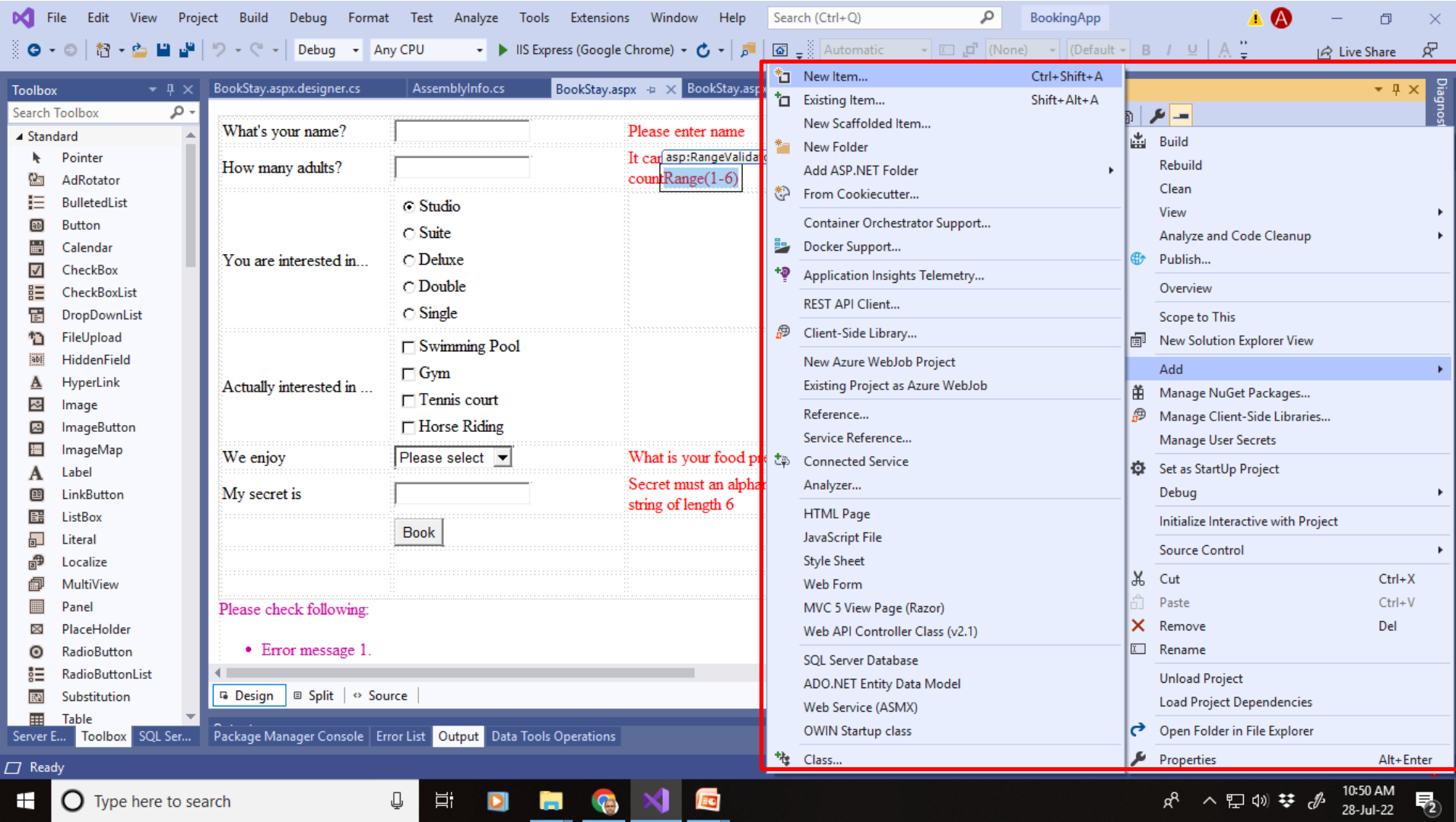
10:47 AM 28-Jul-22

Let's start...

The screenshot displays the Visual Studio IDE with a web application project named "BookingApp". The main window shows the "Design" view of the "BookStay.aspx" page. The page contains a form with the following elements:

- Name Field:** A text input field with the label "What's your name?". A red validation message "Please enter name" is displayed next to it.
- Adults Field:** A text input field with the label "How many adults?". A red validation message "It can't be less than 1" is displayed next to it.
- Interests Field:** A group box labeled "You are interested in..." containing radio buttons for "Studio", "Suite", "Deluxe", "Double", and "Single".
- Facilities Field:** A group box labeled "Actually interested in ..." containing checkboxes for "Swimming Pool", "Gym", "Tennis court", and "Horse Riding".
- Food Preference Field:** A dropdown menu with the label "We enjoy". A red validation message "What is your food preference?" is displayed next to it.
- Secret Field:** A text input field with the label "My secret is". A red validation message "Secret must be an alphanumeric string of length 6" is displayed next to it.
- Buttons:** A "Book" button is located at the bottom of the form.
- Validation Messages:** A pink message "Please check following:" is displayed, followed by a red message "Error message 1."

The Solution Explorer on the right shows the project structure, including the "BookingApp" project, "AssemblyInfo.cs", "BookStay.aspx", "BookStay.aspx.cs", "BookStay.aspx.designer.cs", "packages.config", and "Web.config". The Properties window shows the "BookingApp" project properties, including "Always Start When Debugging", "Anonymous Authentication", "Managed Pipeline Mode", "Project File", "Project Folder", "SSL Enabled", "SSL URL", "URL", and "Windows Authentication".



Visual Studio interface showing the "Add New Item - BookingApp" dialog box. The dialog is used to add a new item to the project.

Toolbox (Left): Standard controls including Pointer, AdRotator, BulletedList, Button, Calendar, CheckBox, CheckBoxList, DropDownList, FileUpload, HiddenField, HyperLink, Image, ImageButton, ImageMap, Label, LinkButton, ListBox, Literal, Localize, MultiView, Panel, Placeholder, RadioButton, RadioButtonList, Substitution, and Table.

Add New Item - BookingApp Dialog:

- Installed:**
 - Visual C#
 - Code
 - Data
 - General
 - Web
 - General
 - Markup
 - Scripts
 - Web Forms
 - MVC
 - Razor
 - SignalR
 - Web API
 - Windows Forms
 - WPF
 - Extensibility
 - SQL Server
 - Storm Items
 - Online
- Sort by:** Default
- Search (Ctrl+E):** (Empty search bar)
- Items:**
 - SignalR Persistent Connection Class (v2) Visual C#
 - ASP.NET Handler Visual C#
 - ASP.NET Module Visual C#
 - Browser File Visual C#
 - Dynamic Data Field Visual C#
 - Generic Handler Visual C#
 - Global Application Class Visual C#** (Selected)
 - JavaScript JSON Configuration File Visual C#
 - JSON File Visual C#
 - JSON Schema File Visual C#
 - JSX File Visual C#
 - LESS Style Sheet Visual C#
- Name:** Global.aspx (Entered)
- Buttons:** Add, Cancel

Always Start when Debugging: Start the local Web server even when not the startup project

Windows Taskbar: Ready, Type here to search, 10:50 AM, 28-Jul-22

Visual Studio interface showing the code editor for `Global.asax.cs` in the `BookingApp` project. The code defines the `Global` class, which inherits from `System.Web.HttpApplication`, and implements several application events.

```
10 public class Global : System.Web.HttpApplication
11 {
12
13     protected void Application_Start(object sender, EventArgs e)
14     {
15
16     }
17
18     protected void Session_Start(object sender, EventArgs e)
19     {
20
21     }
22
23     protected void Application_BeginRequest(object sender, EventArgs e)
24     {
25
26     }
27
28     protected void Application_AuthenticateRequest(object sender, EventArgs e)
29     {
30
31     }
32
33     protected void Application_Error(object sender, EventArgs e)
```

The right sidebar displays the **Solution Explorer** showing the project structure for `BookingApp`:

- Solution 'BookingApp' (1 of 1 project)
 - BookingApp
 - Connected Services
 - Properties
 - AssemblyInfo.cs
 - References
 - BookStay.aspx
 - BookStay.aspx.cs
 - BookStay.aspx.designer.cs
 - Global.asax
 - packages.config
 - Web.config

The **Properties** window is also visible, showing the selected item's properties.

The status bar at the bottom indicates "No issues found" and shows the current line and column (Ln: 1, Ch: 1).

Global.asax

- Also known as the ASP.NET application file
- An optional file
- It allows us to write event handlers that react to global events in web applications.
- Used to handle server side events
 - application level events
- Global.asax file is never called directly by the user, rather they are called automatically in response to application events.
- Resides in root directory
- A website can have only one Global.asax file
- During execution, Global.asax is parsed and compiled into a dynamically generated .NET framework class derived from `HttpApplication`

Responding to Application Events

- `HttpApplication` class
 - Provides several events that you can handle to perform actions when ASP.NET raises certain events at application level
 - Initializing variables values when your application starts
 - Logging requests to your application
 - Handling application level errors
 - Etc...

Events

- `Application_Start()`
 - Start event is raised when your application is started by IIS
 - Useful for initializing variables that are scoped at the application level
- `Application_End()`
 - End event is raised when your application stops or shuts down
 - Useful to free application-level resources or perform some sort of logging
- `Application_Error()`
 - Error event is raised when an unhandled error occurs and rises up to the application scope
 - Useful to perform worst-case, catch all error logging

Events

- `Session_Start()`
 - This event raised for each time a new session begins, This is a good place to put code that is session-specific
- `Session_End()`
 - This event called when session of user ends

Visual Studio interface showing the code for `BookingApp.Global` in `Global.asax.cs`. The code defines the `Global` class, which inherits from `System.Web.HttpApplication`. It implements the `Application_Start`, `Session_Start`, `Application_BeginRequest`, and `Application_AuthenticateRequest` methods.

```
10 public class Global : System.Web.HttpApplication
11 {
12
13     protected void Application_Start(object sender, EventArgs e)
14     {
15         Application["UsersOnline"] = 0;
16     }
17
18     protected void Session_Start(object sender, EventArgs e)
19     {
20         Application.Lock();
21         Application["UsersOnline"] = (int)Application["UsersOnline"] + 1;
22         Session["UserName"] = "Guest";
23         Application.Unlock();
24     }
25
26     protected void Application_BeginRequest(object sender, EventArgs e)
27     {
28     }
29
30
31     protected void Application_AuthenticateRequest(object sender, EventArgs e)
32     {
33 }
```

Why are the code within lock and unlock?

The `Application.Lock()` and `Application.Unlock()` methods are used to ensure thread safety when accessing the `Application` object's state (e.g., `UsersOnline`). In a multi-threaded environment, multiple requests can simultaneously access and modify the state, leading to race conditions or data corruption. Locking the application object ensures that only one thread can execute the code within the `Session_Start` method at a time, preventing such issues.

The interface also shows the Solution Explorer on the right, displaying the project structure for `BookingApp`, including `AssemblyInfo.cs`, `BookStay.aspx`, `BookStay.aspx.designer.cs`, `Global.asax`, `packages.config`, and `Web.config`.

File Edit View Project Build Debug Test Analyze Tools Extensions Window Help Search (Ctrl+Q) BookingApp

Debug Any CPU IIS Express (Google Chrome)

Toolbox Search Tool: General

There are no usable controls in this group. Drag an item onto this text to add it to the toolbox.

Web.config Global.asax.cs BookStay.aspx.designer.cs AssemblyInfo.cs BookStay.aspx BookStay.aspx.cs

BookingApp BookingApp.Global Application_End(object sender, EventArgs e)

```
32 {
33 }
34 }
35
36 protected void Application_Error(object sender, EventArgs e)
37 {
38 }
39 }
40
41 protected void Session_End(object sender, EventArgs e)
42 {
43     Application.Lock();
44     Application["UsersOnline"] = (int)Application["UserOnline"] - 1;
45     Application.UnLock();
46 }
47
48 protected void Application_End(object sender, EventArgs e)
49 {
50 }
51 }
52 }
53 }
```

129 % No issues found Ln: 51 Ch: 10 SPC CRLF

Package Manager Console Error List Output Data Tools Operations

Solution Explorer

Search Solution Explorer (Ctrl+;)

Solution 'BookingApp' (1 of 1 project)

- BookingApp
 - Connected Services
 - Properties
 - AssemblyInfo.cs
 - References
 - BookStay.aspx
 - BookStay.aspx.cs
 - BookStay.aspx.designer.cs
 - Global.asax
 - Global.asax
 - packages.config
 - Web.config

Properties

Item(s) Saved

Type here to search

11:05 AM 28-Jul-22

Visual Studio interface showing the development of a web application named BookingApp.

Menu Bar: File, Edit, View, Project, Build, Debug, Test, Analyze, Tools, Extensions, Window, Help.

Search Bar: Search (Ctrl+Q)

Toolbox: Search Tool: [magnifying glass icon]. General tab. Message: "There are no usable controls in this group. Drag an item onto this text to add it to the toolbox."

Web.config:

```
1 <system>
2   <system.collections.generic>
3   <system.linq>
4   <system.web>
5   <system.web.ui>
6   <system.web.ui.webcontrols>
7
8 </system>
9
10 <page>
11   <public partial class BookStay : System.Web.UI.Page
12     <protected void Page_Load(object sender, EventArgs e)
13     {
14       this.UnobtrusiveValidationMode = System.Web.UI.UnobtrusiveValidationMode.None;
15
16       Response.Write("Users count:" + (int)Application["UsersOnline"] + "<br/>");
17       Response.Write("Session name: " + Session["UserName"] + "<br/>");
18     }
19   }
20
21   <protected void btnSubmit_Click(object sender, EventArgs e)
22   {
23     string name = tbName.Text;
24     Session["UserName"] = name;
```

BookStay.aspx.designer.cs:

```
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```

AssemblyInfo.cs:

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```

BookStay.aspx:

```
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21
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23
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```

BookStay.aspx.cs:

```
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84
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86
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88
89
90
91
92
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99
100
```

Solution Explorer: Solution 'BookingApp' (1 of 1 project). BookingApp. Properties. AssemblyInfo.cs. References. BookStay.aspx. BookStay.aspx.cs. BookStay.aspx.designer.cs. Global.asax. packages.config. Web.config.

Properties: [Properties window content]

Package Manager Console: [Package Manager Console content]

Error List: [Error List content]

Output: [Output content]

Data Tools Operations: [Data Tools Operations content]

Status Bar: 129% | No issues found | Ln: 19 Ch: 10 SPC CRLF

Taskbar: Windows Start button, Search bar (Type here to search), Task View, File Explorer, Google Chrome, Visual Studio, Teams, OneDrive.

System Tray: Add to Source Control, 11:05 AM, 28-Jul-22, Notification area (2 icons).

File Edit View Project Build Debug Test Analyze Tools Extensions Window Help Search (Ctrl+Q) BookingApp

Debug Any CPU IIS Express (Google Chrome)

Toolbox Search Tool: General

There are no usable controls in this group. Drag an item onto this text to add it to the toolbox.

Web.config Global.asax.cs BookStay.aspx.designer.cs AssemblyInfo.cs BookStay.aspx BookStay.aspx.cs btnSubmit_Click(object sender, EventArgs e)

```
17 Response.Write("Session name: " + Session["UserName"] + "<br/>");
18
19 }
20
21 protected void btnSubmit_Click(object sender, EventArgs e)
22 {
23     string name = tbName.Text;
24     Session["UserName"] = name;
25     int pax = Int32.Parse(tbACount.Text);
26     string roomType = rblistRoomType.SelectedItem.Value;
27     List<ListItem> opFeat = new List<ListItem>();
28     foreach (ListItem item in cklistOpFacility.Items)
29     {
30         if (item.Selected)
31         {
32             opFeat.Add(item);
33         }
34     }
35     string foodPref;
36     if (ddlFoodPref.SelectedItem.Value == "0")
37     {
38         foodPref = "you have not selected food preference. Kindly oblige.";
39     }
40     else
```

129 % No issues found Ln: 24 Ch: 13 SPC CRLF

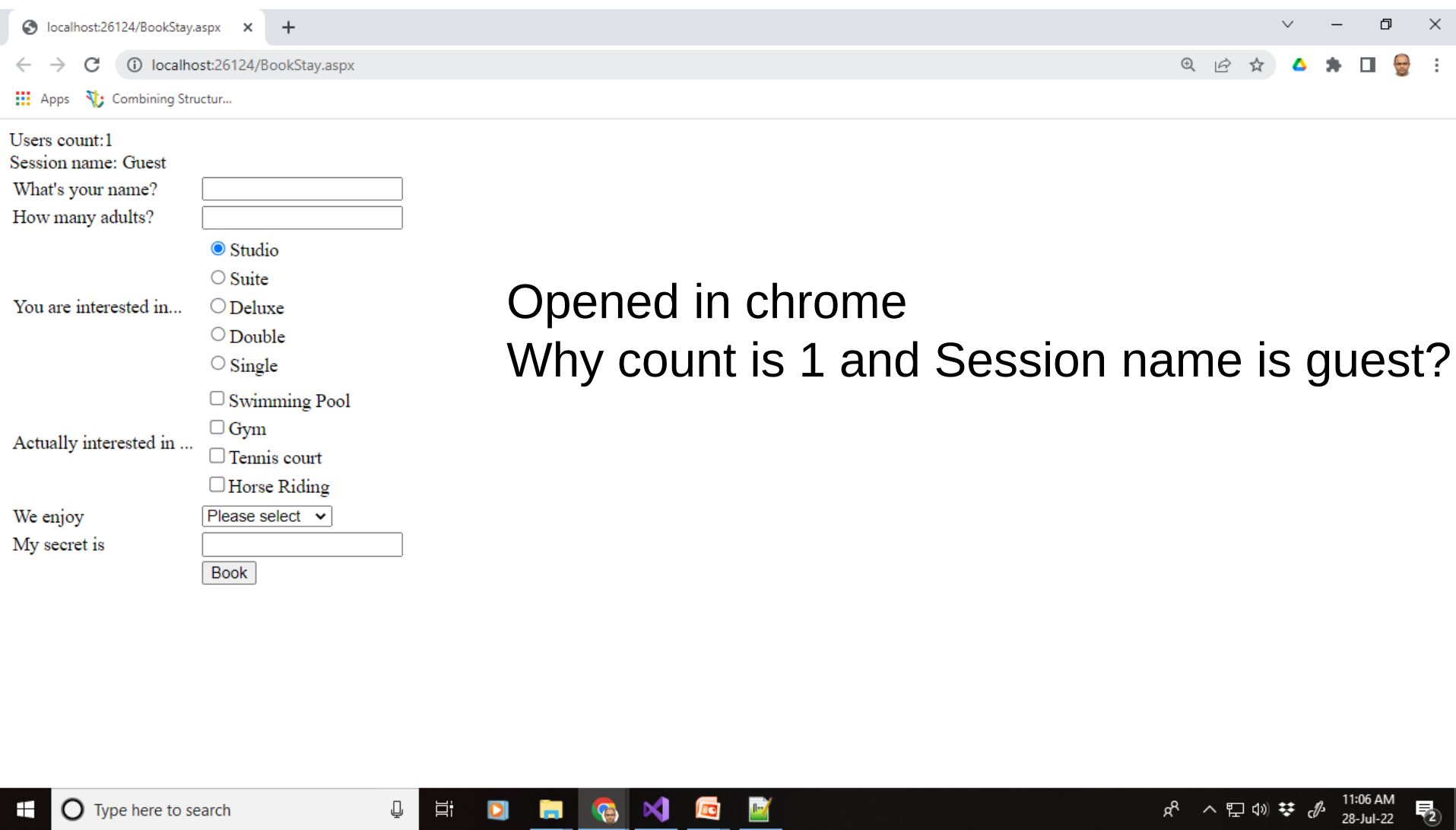
Package Manager Console Error List Output Data Tools Operations

Solution Explorer Search Solution Explorer (Ctrl+;) Solution 'BookingApp' (1 of 1 project) BookingApp Connected Services Properties AssemblyInfo.cs References BookStay.aspx BookStay.aspx.cs BookStay.aspx.designer.cs Global.asax packages.config Web.config

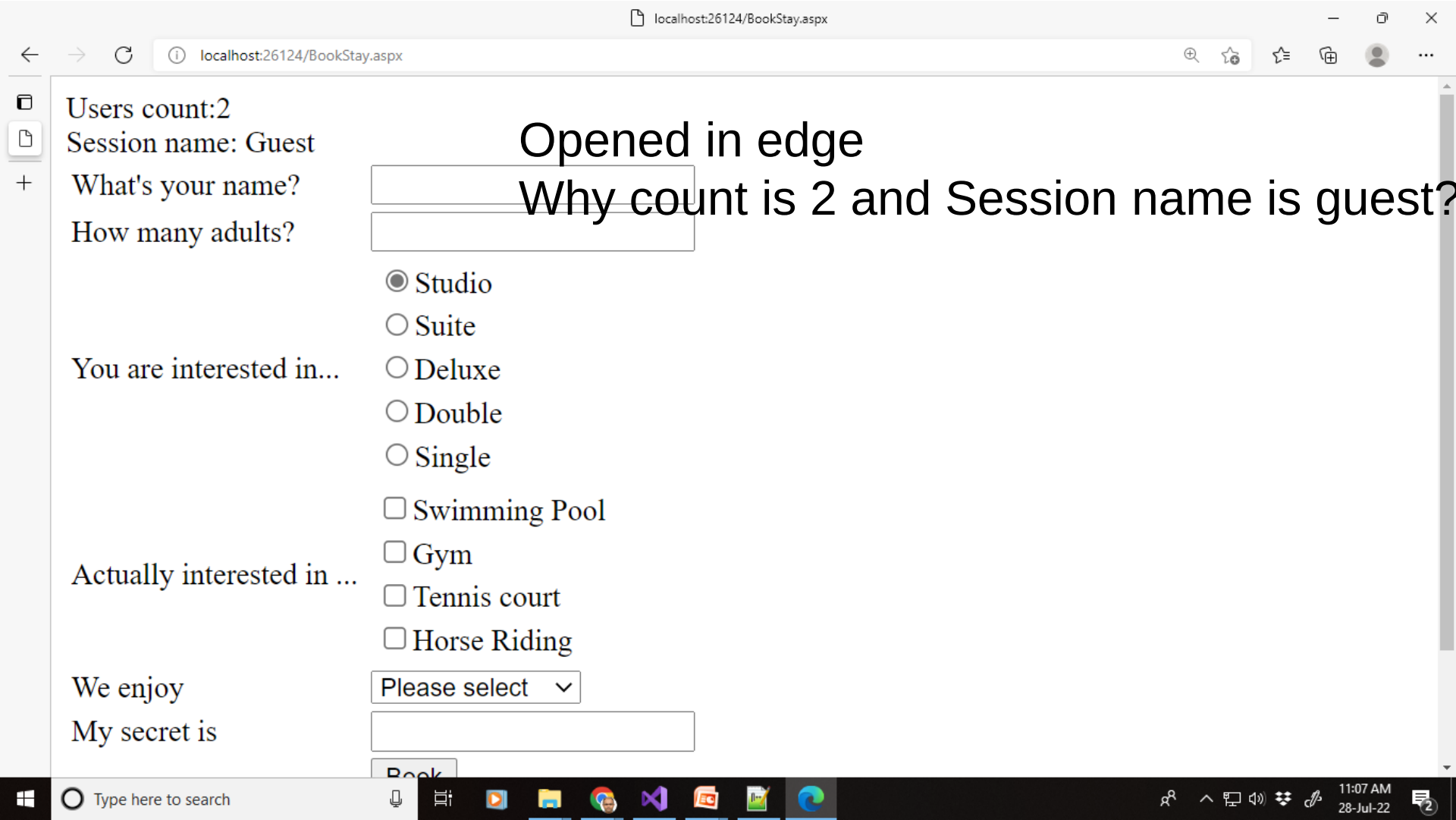
Properties

Item(s) Saved Add to Source Control

Type here to search 11:05 AM 28-Jul-22

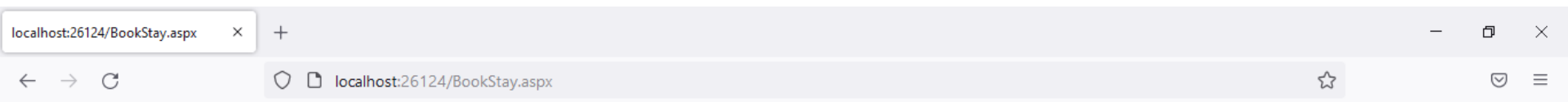


Opened in chrome
Why count is 1 and Session name is guest?



Opened in edge

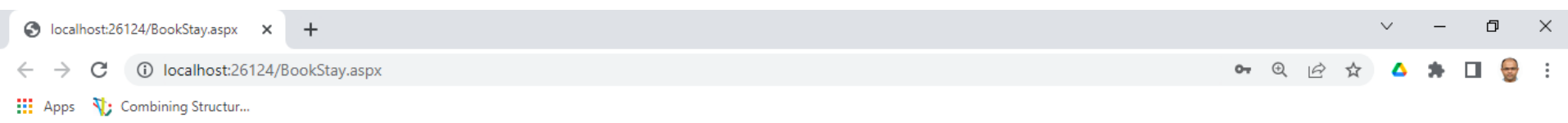
Why count is 2 and Session name is guest?



Users count:3
Session name: Guest
What's your name?
How many adults?
☒ Studio
☐ Suite
You are interested in... ☐ Deluxe
☐ Double
☐ Single
☐ Swimming Pool
Actually interested in ... ☐ Gym
☐ Tennis court
☐ Horse Riding
We enjoy
My secret is

Opened in firefox
Why count is 3 and Session name is guest?





Fill the form and click Book

Users count: 1
Session name: Guest

What's your name?

How many adults?

You are interested in...

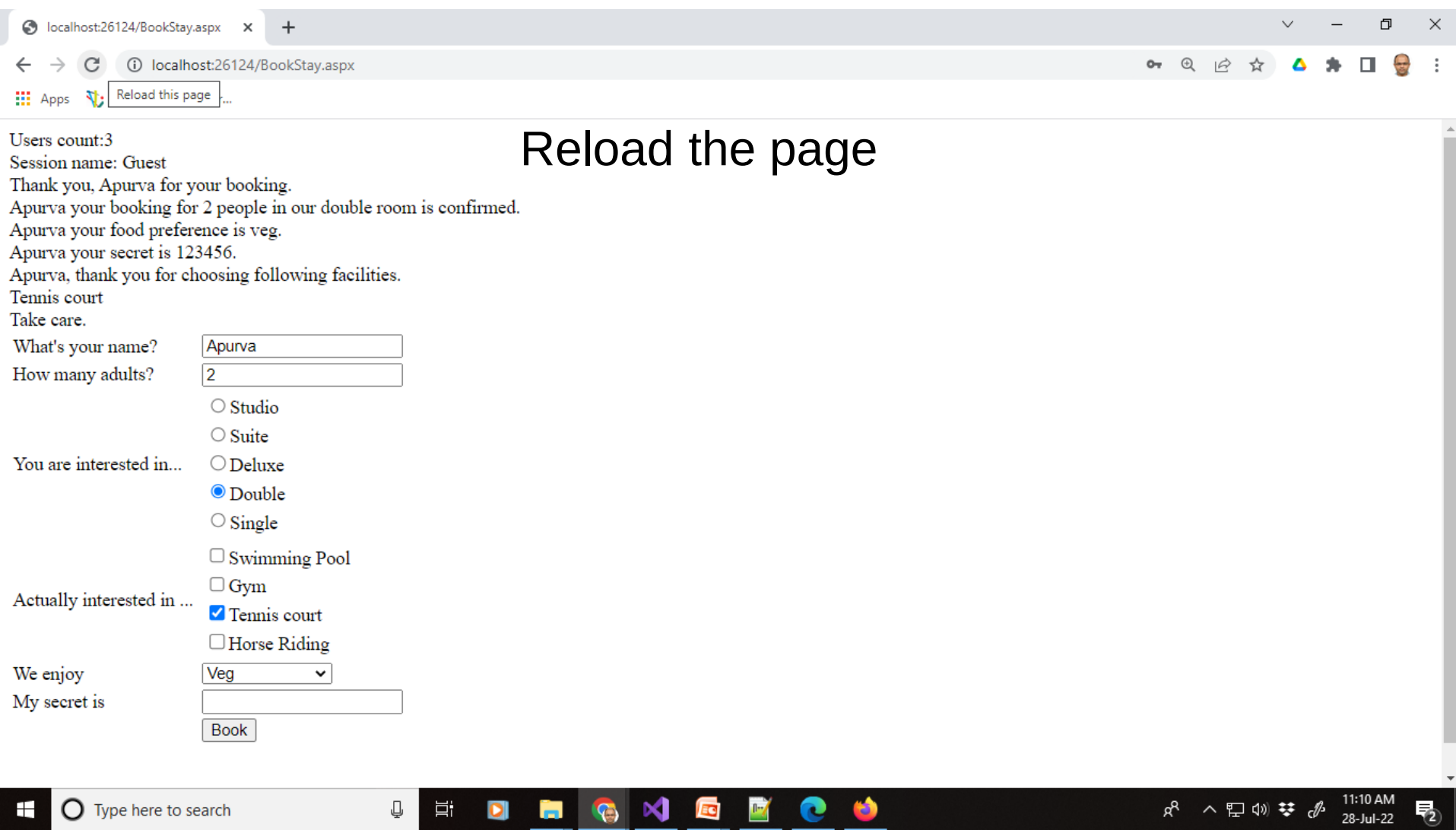
☐ Studio
☐ Suite
☐ Deluxe
☒ Double
☐ Single

Actually interested in ...

☐ Swimming Pool
☐ Gym
☒ Tennis court
☐ Horse Riding

We enjoy

My secret is



Why count is 3 and Session name is Apurva?

Users count:3
Session name: Apurva
Thank you, Apurva for your booking.
Apurva your booking for 2 people in our double room is confirmed.
Apurva your food preference is veg.
Apurva your secret is 123456.
Apurva, thank you for choosing following facilities.
Tennis court
Take care.

What's your name?

How many adults?

You are interested in...

- ☐ Studio
- ☐ Suite
- ☐ Deluxe
- ☒ Double
- ☐ Single

Actually interested in ...

- ☐ Swimming Pool
- ☐ Gym
- ☒ Tennis court
- ☐ Horse Riding

We enjoy

My secret is

File Edit View Project Build Debug XML Test Analyze Tools Extensions Window Help Search (Ctrl+Q) BookingApp

Process: [11844] iisexpress.exe Lifecycle Events Thread: Stack Frame:

Server Explorer

- Azure
- Data Connections
- Servers

Web.config

```
1 <?xml version="1.0" encoding="utf-8"?>
2
3 <!--
4 For more information on how to configure your ASP.NET application, please visit
5 https://go.microsoft.com/fwlink/?LinkId=169433
6 -->
7 <configuration>
8   <system.web>
9     <sessionState mode="InProc" timeout="10"></sessionState>
10    <compilation debug="true" targetFramework="4.7.2"/>
11    <httpRuntime targetFramework="4.7.2"/>
12  </system.web>
13  <system.codedom>
14    <compilers>
15      <compiler language="c#;cs;csharp" extension=".cs"
16        type="Microsoft.CodeDom.Providers.DotNetCompilerPlatform.CSharpCodeProvider, Microsoft.CodeDom.Providers.DotNetCompilerPlatform, Version=4.7.2.0, Culture=neutral, PublicKeyToken=b03f567c3d87aa3f, processorArchitecture=MSIL"/>
17    </compilers>
18  </system.codedom>
19 </configuration>
```

What if timeout value is 1?

129% No issues found Ln: 12 Ch: 16 SPC CRLF

Autos

Search (Ctrl+E) Search Depth: 3

Name	Value	Type
------	-------	------

Autos Locals Watch 1

Error List

Entire Solution 0 Errors

Search Error List

Code	Description	Project
IDE1006	Naming rule violation: These words must begin with upper case characters:	BookingApp

Call S... Break... Excep... Com... Imme... Output Error...

Ready

Type here to search

11:11 AM 28-Jul-22

Visual Studio interface showing the development of a web application named **BookingApp**. The main editor displays the code for **BookStay.aspx.cs**, specifically the **btnSubmit_Click** event handler. A red arrow points to the line **ViewState["pax"] = pax;**, which is highlighted in blue.

```
14 this.UnobtrusiveValidationMode = System.Web.UI.UnobtrusiveValidationMode.None;
15
16 Response.Write("Users count:" + (int)Application["UsersOnline"] + "<br/>");
17 Response.Write("Session name: " + Session["UserName"] + "<br/>");
18
19 }
20
21 protected void btnSubmit_Click(object sender, EventArgs e)
22 {
23     string name = tbName.Text;
24     Session["UserName"] = name;
25     int pax = Int32.Parse(tbACount.Text);
26     ViewState["pax"] = pax;
27     string roomType = rblistRoomType.SelectedItem.Value;
28     List<ListItem> opFeat = new List<ListItem>();
29     foreach (ListItem item in cklistOpFacility.Items)
30     {
31         if (item.Selected)
32         {
33             opFeat.Add(item);
34         }
35     }
36     string foodPref;
37     if (ddlFoodPref.SelectedItem.Value == "0")
```

The Solution Explorer on the right shows the project structure:

- BookingApp
 - Connected Services
 - Properties
 - AssemblyInfo.cs
 - References
 - BookStay.aspx
 - BookStay.aspx.cs
 - BookStay.aspx.designer.cs
 - CheckViewState.aspx
 - CheckViewState.aspx.cs
 - CheckViewState.aspx.designer.cs

The Properties window is empty.

At the bottom, the status bar shows "129 %", "No issues found", and "Ln: 26 Ch: 36 SPC CRLF". The Windows taskbar at the very bottom shows the time as 1:45 PM on 28-Jul-22.

Visual Studio interface showing the code editor for `BookStay.aspx.cs` in the `BookingApp` project. The code is for the `btnSubmit_Click` event handler. A red arrow points to the line `Response.Write($"ViewState:<b>{ViewState["pax"]}</b><br/>");`.

```
44 }
45 string secret = tbPassword.Text;
46 Response.Write($"Thank you, {name} for your booking.<br/>");
47 Response.Write($"{{name}} your booking for {{pax}} people in " +
48     $"our {{roomType}} room is confirmed. <br/>");
49 Response.Write($"{{name}} your food preference is {{foodPref}}. <br/>");
50 Response.Write($"{{name}} your secret is {{tbPassword.Text}}. <br/>");
51 Response.Write($"{{name}}, thank you for choosing following facilities. <br/>");
52 foreach (ListItem temp in opFeat){
53     Response.Write(temp.Text + "<br/>");
54 }
55 Response.Write("Take care.<br/>");
56 Response.Write($"ViewState:<b>{ViewState["pax"]}</b><br/>");
57 }
58 }
59 }
60 }
61 }
```

The Solution Explorer on the right shows the project structure:

- BookingApp
 - Connected Services
 - Properties
 - AssemblyInfo.cs
 - References
 - BookStay.aspx
 - BookStay.aspx.cs
 - BookStay.aspx.designer.cs
 - CheckViewState.aspx
 - CheckViewState.aspx.cs
 - CheckViewState.aspx.designer.cs

The Properties window is empty.

Bottom status bar: 129% zoom, No issues found, Ln: 56, Ch: 73, SPC, CRLF.

Windows taskbar at the bottom shows the Start button, search bar, and taskbar icons for File Explorer, Google Chrome, Visual Studio, and a calendar icon.

Visual Studio interface showing a web application named "BookingApp" running in IIS Express (Google Chrome). The application is a form for booking a stay, with fields for name, number of adults, interests, and a secret. The form includes validation messages and a "Book" button.

The form fields and their associated validation messages are:

- What's your name? (Text input) - Please enter name
- How many adults? (Text input) - It can't be 0Please enter a countRange(1-6)
- You are interested in... (Radio buttons):
 - Studio
 - Suite
 - Deluxe
 - Double
 - Single
- Actually interested in ... (Checkboxes):
 - Swimming Pool
 - Gym
 - Tennis court
 - Horse Riding
- We enjoy (Dropdown menu) - Please select
- My secret is (Text input) - What is your food preference?
Secret must an alphanumeric string of length 6

The form also includes a "Book" button and a link labeled "Can ViewState travel?" with the text "Please check following:" and two error messages:

- Error message 1.
- Error message 2.

The Solution Explorer shows the "hableViewState" property of the "System.Web.UI.WebControls.HyperLink" control. The properties list includes:

Property	Value
(Expressions)	
(ID)	hableViewState
AccessKey	
BackColor	
BorderColor	
BorderStyle	NotSet
BorderWidth	
ClientIDMode	Inherit
CssClass	
Enabled	True
EnableTheming	True
EnableViewState	True
Font	
ForeColor	
Height	
ImageHeight	
ImageUrl	
ImageWidth	
NavigateUrl	~/CheckViewState.aspx
SkinID	
TabIndex	0
Target	
Text	Can ViewState travel?
ToolTip	

The "Text" property is highlighted in blue, and the "Can ViewState travel?" text is visible in the design view. A red arrow points from the "Can ViewState travel?" text in the design view to the "Text" property in the Properties window.

Users count: 1
Session name: Guest

What's your name?

How many adults?

☐ Studio
☐ Suite

You are interested in... ☐ Deluxe
☒ Double
☐ Single

☒ Swimming Pool

Actually interested in ... ☐ Gym
☐ Tennis court
☐ Horse Riding

We enjoy

My secret is

[Can ViewState travel?](#)

localhost:26124/BookStay.aspx

localhost:26124/BookStay.aspx

AppsCombining Structur...

Thank you, Apurva for your booking.

Apurva your booking for 2 people in our double room is confirmed.

Apurva your food preference is veg.

Apurva your secret is 123456.

Apurva, thank you for choosing following facilities.

Swimming Pool

Take care.

ViewState:2

What's your name?

Apurva

How many adults?

2

You are interested in...

☐ Studio

☐ Suite

☐ Deluxe

☒ Double

☐ Single

Actually interested in ...

☒ Swimming Pool

☐ Gym

☐ Tennis court

☐ Horse Riding

We enjoy

Veg

My secret is

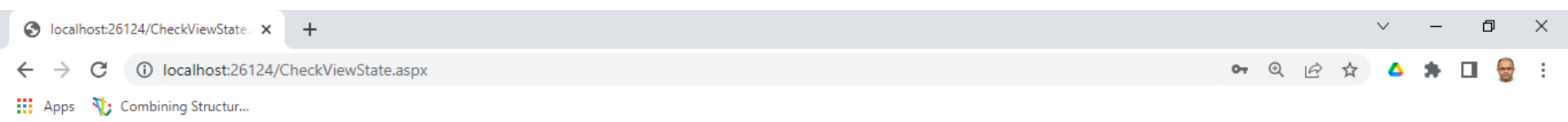
Book

Can ViewState travel?

Click

Type here to search

1:50 PM28-Jul-22



The ViewState can not travel from one page to another page.
You may try time travel.



So, View state is **used automatically by the ASP.NET page framework to persist information that must be preserved between postbacks.** This information includes any non-default values of controls. You can also use view state to store application data that is specific to a page.

Is the ViewState useful?

Visual Studio interface showing a web application project named "BookingApp". The main window displays the design view of a web page titled "Do you want to check your past?". The page contains a radio button group with two options: "yes" and "no". Below the radio buttons is a "Sure" button. The "yes" radio button is selected.

The Solution Explorer on the right shows the project structure, including files like "CheckViewState.aspx", "AssemblyInfo.cs", "BookStay.aspx.cs", "BookStay.aspx", "CheckViewState.aspx.cs", and "Global.aspx.cs". The Properties window on the right shows the properties of the selected control, "rbChoice" (System.Web.UI.WebControls.RadioButton). The "DataSourceID" property is highlighted.

Properties	
rbChoice System.Web.UI.WebControls.RadioButton	
(Expressions)	
(ID)	rbChoice
AccessKey	
AppendDataBoundItem	False
AutoPostBack	False
BackColor	
BorderColor	
BorderStyle	NotSet
BorderWidth	
CausesValidation	False
CellPadding	-1
CellSpacing	-1
ClientIDMode	Inherit
CssClass	
DataMember	
DataSourceID	
DataTextField	
DataTextFormatString	
DataValueField	
Enabled	True
EnableTheming	True
EnableViewState	True
Font	
ForeColor	

DataSourceID
The control ID of an IDDataSource that will be used as the data source.

Ready | Type here to search | 2:11 PM 28-Jul-22

Visual Studio interface showing a web application project named "BookingApp". The main window displays the design view of a web page with a form titled "Do you want to check your past?". The form contains a radio button list control named "asp:radiobuttonlist#rbChoice" with two options: "yes" and "no".

The "ListItem Collection Editor" dialog is open, showing the "yes" property selected. The "Members" list contains two items: "0 yes" and "1 no". The "yes properties" section shows the following values:

yes properties:	
Misc	
Enabled	True
Selected	False
Text	yes
Value	yes

The "Properties" window on the right shows the properties for the "rbChoice" control (System.Web.UI.WebControls.RadioButtonList). The "Data Source ID" property is highlighted, indicating it is the control ID of an IDataSource that will be used as the data source.

The bottom status bar shows the time as 2:11 PM on 28-Jul-22.

Visual Studio interface showing a web application project named "BookingApp". The main window displays the design view of a web page with the text "Do you want to check your past?" and a button labeled "Sure". The button's ID is "asp:button#btnSubmit".

The Solution Explorer on the right shows the project structure, including the "btnSubmit" button control. The Properties window displays the properties for the "btnSubmit" button, including:

- (Expressions)
- (ID) **btnSubmit**
- AccessKey
- BackColor
- BorderColor
- BorderStyle **NotSet**
- BorderWidth
- CausesValidation **True**
- ClientIDMode **Inherit**
- CommandArgument
- CommandName
- CssClass
- Enabled **True**
- EnableTheming **True**
- EnableViewState **True**
- Font
- ForeColor
- Height
- OnClick
- PostBackUrl
- SkinID
- TabIndex **0**
- Text** **Sure**
- ToolTip

The Text property is highlighted, showing the text "Sure".

The bottom status bar indicates the application is "Ready". The Windows taskbar at the bottom shows the time as 2:11 PM on 28-Jul-22.

Visual Studio interface showing a web application project named "BookingApp". The main window displays the design view of "CheckViewState.aspx", which contains a form with a label "Do you want to check your past?" and a button labeled "Sure". The button's ID is "asp:button#btnSubmit".

The Solution Explorer on the right shows the project structure, including "btnSubmit" (System.Web.UI.WebControls.Button). The Properties window displays the "Click" event, which is set to "btnSubmit_Click".

The Properties window also shows the "Click" event description: "Fires when the button is clicked."

The bottom status bar indicates the application is "Ready".

Visual Studio interface showing the code for `CheckViewState.aspx.cs` in the `BookingApp` project.

The code defines a protected method `btnSubmit_Click` that handles the click event of a button. It checks the selected item of a radio button (`rbChoice`). If the selected item is "yes", it constructs a URL pointing to `BookStay.aspx` with query parameters `que="Do you know?"` and `ans` (URL-encoded text of the selected item), and redirects the response to that URL. Otherwise, it redirects the response to `~/CheckViewState.aspx`.

```
22 }
23 }
24
25 protected void btnSubmit_Click(object sender, EventArgs e)
26 {
27     if(rbChoice.SelectedItem.Text == "yes")
28     {
29         string URL = "BookStay.aspx?";
30         URL += "que=" + "Do you know?" + "&";
31         URL += "ans=" + Server.UrlEncode(rbChoice.SelectedItem.Text) + "&";
32         Response.Redirect(URL);
33     }
34     else
35     {
36         Response.Redirect("~/CheckViewState.aspx");
37     }
38 }
39
40
```

The interface includes the following components:

- Menu Bar:** File, Edit, View, Project, Build, Debug, Test, Analyze, Tools, Extensions, Window, Help.
- Search Bar:** Search (Ctrl+Q).
- Toolbar:** Includes buttons for running, debugging, and other development actions.
- Toolbox:** On the left, it shows a message: "There are no usable controls in this group. Drag an item onto this text to add it to the toolbox."
- Solution Explorer:** Displays the project structure, including `BookingApp` and its sub-items.
- Properties Window:** On the right, it shows the properties of the selected item.
- Diagnostic Tools:** On the far right, it provides a place for diagnostic information.
- Status Bar:** At the bottom, it shows the current file (`CheckViewState.aspx.cs`), line number (40), and character (2).

File Edit View Project Build Debug Test Analyze Tools Extensions Window Help Search (Ctrl+Q) BookingApp

Debug Any CPU IIS Express (Google Chrome)

Toolbox Search Tool: BookingApp BookingApp.BookStay Page_Load(object sender, EventArgs e)

General

There are no usable controls in this group. Drag an item onto this text to add it to the toolbox.

```
7
8 namespace BookingApp
9 {
10     public partial class BookStay : System.Web.UI.Page
11     {
12         protected void Page_Load(object sender, EventArgs e)
13         {
14             this.UnobtrusiveValidationMode = System.Web.UI.UnobtrusiveValidationMode.None;
15
16             Response.Write("Users count:" + (int)Application["UsersOnline"] + "<br/>");
17             Response.Write("Session name: " + Session["UserName"] + "<br/>");
18
19             if(Request.QueryString["que"]!=null) Why is this check?
20             {
21                 string q = Request.QueryString["que"];
22                 string a = Request.QueryString["ans"];
23                 Response.Write($"Your question was: {q} and your answer was: {a}.");
24             }
25
26         }
27
28         protected void btnSubmit_Click(object sender, EventArgs e)
29         {
30             string name = tbName.Text;
```

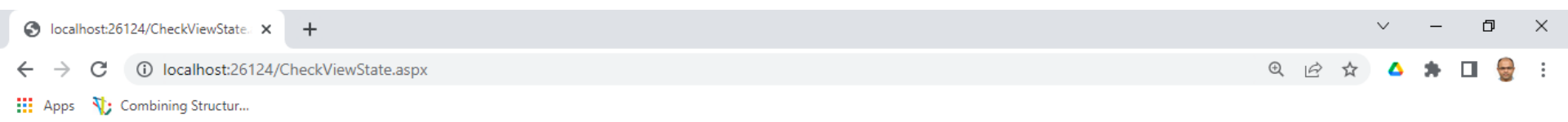
129 % No issues found Ln: 26 Ch: 10 SPC CRLF

Package Manager Console Error List Output Data Tools Operations

Ready Add to Source Control

Type here to search

2:12 PM 28-Jul-22



The ViewState can not travel from one page to another page.

You may try time travel.

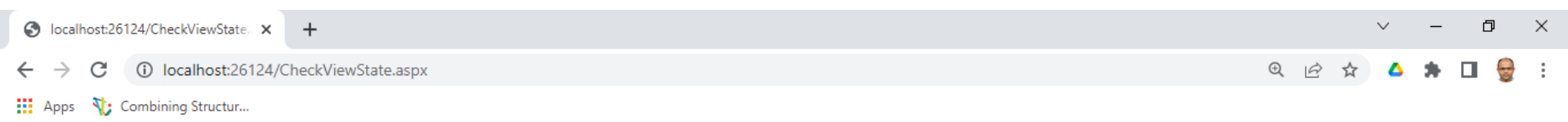
Do you want to check your past?

☐ yes

☒ no

Type here to search

2:12 PM
28-Jul-22



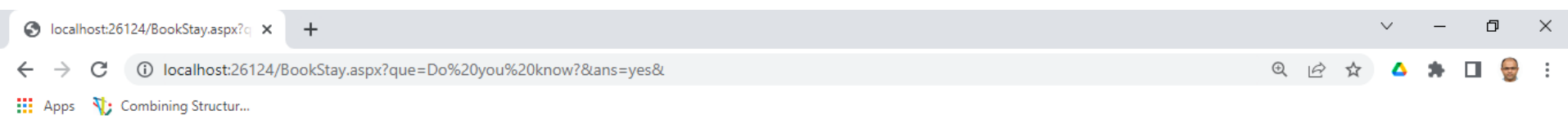
The ViewState can not travel from one page to another page.

You may try time travel.

Do you want to check your past?

☒ yes

☐ no



Users count:1
Session name: Guest
Your question was: Do you know? and your answer was: yes.



From, where is this coming?
Falling from stars or URL. Why?

What's your name?

How many adults?

☒ Studio
☐ Suite

You are interested in... ☐ Deluxe
☐ Double
☐ Single

☐ Swimming Pool
☐ Gym

Actually interested in ... ☐ Tennis court
☐ Horse Riding

We enjoy

My secret is

[Can ViewState travel?](#)

Is the Query String approach useful in maintaining state?



Cookie demo application (Implement yourself)



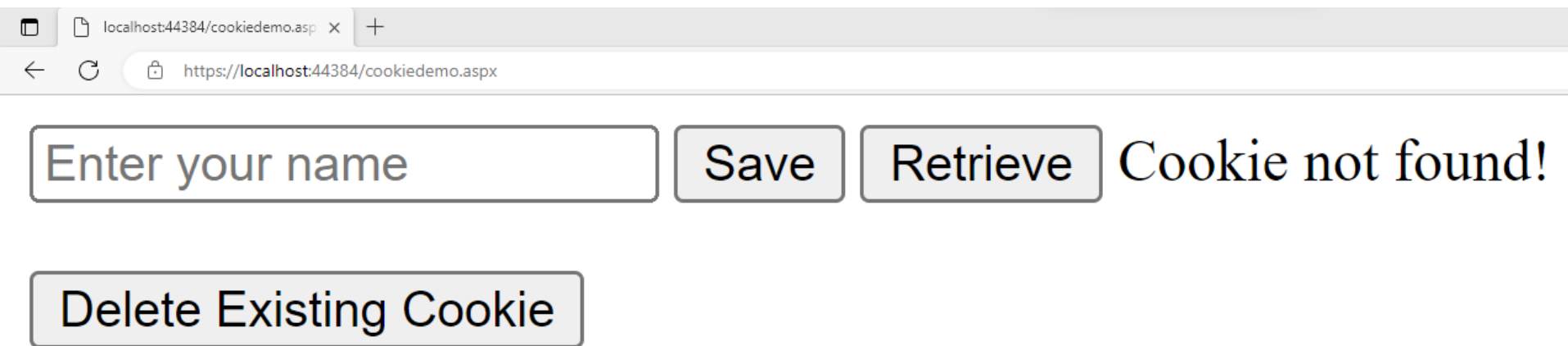
Enter your name

Save

Retrieve

Delete Existing Cookie

When we click retrieve initially



localhost:44384/cookie demo.aspx x +

← ↻ https://localhost:44384/cookie demo.aspx

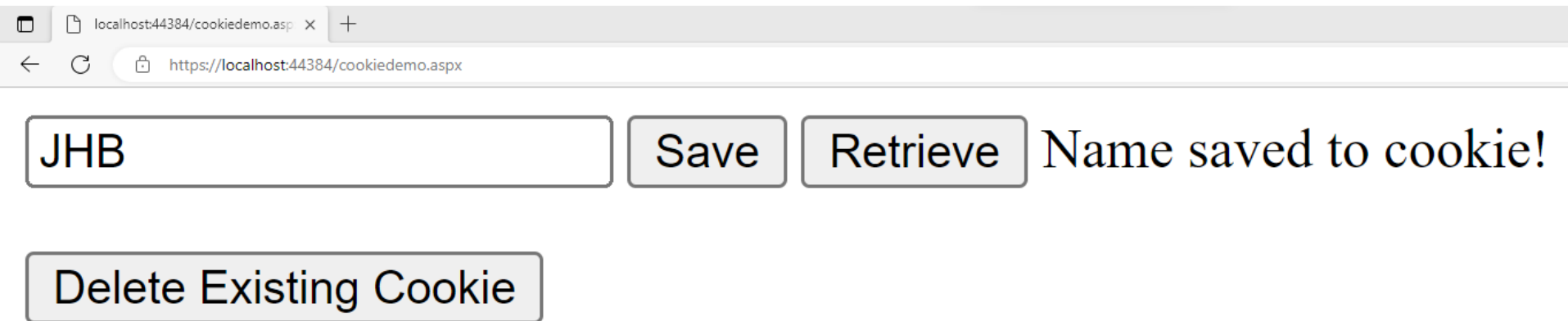
Enter your name

Save Retrieve

Cookie not found!

Delete Existing Cookie

Enter name and click save



The screenshot shows a web browser window with a single tab titled 'localhost:44384/cookie demo.asp'. The address bar displays 'https://localhost:44384/cookie demo.aspx'. The page content includes a text input field containing 'JHB', a 'Save' button, a 'Retrieve' button, and a message 'Name saved to cookie!'. Below these elements is a 'Delete Existing Cookie' button.

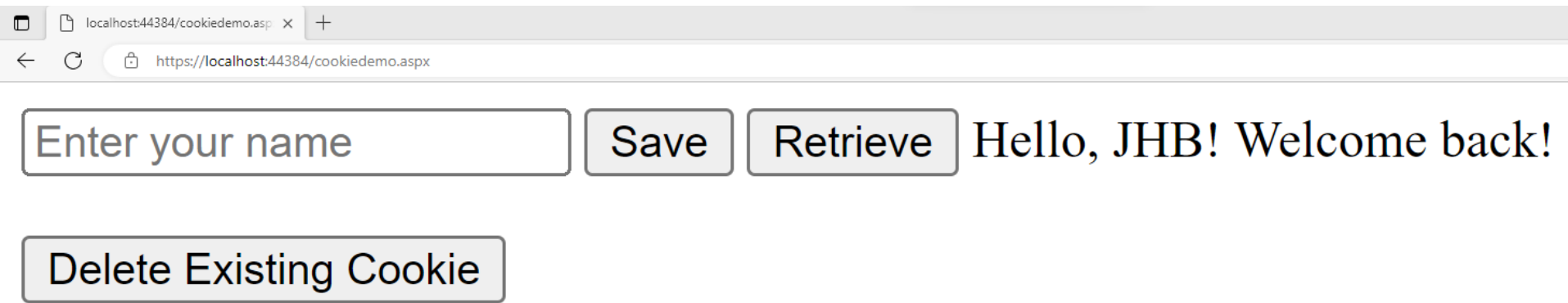
localhost:44384/cookie demo.asp x +

← ↻ https://localhost:44384/cookie demo.aspx

JHB Save Retrieve Name saved to cookie!

Delete Existing Cookie

Close the application, again run and then click retrieve



The screenshot shows a web browser window with a single tab titled 'localhost:44384/cookie demo.asp'. The address bar displays 'https://localhost:44384/cookie demo.aspx'. The page content includes a text input field with the placeholder 'Enter your name', followed by 'Save' and 'Retrieve' buttons. To the right of these buttons is the text 'Hello, JHB! Welcome back!'. Below the input field and buttons is a 'Delete Existing Cookie' button.

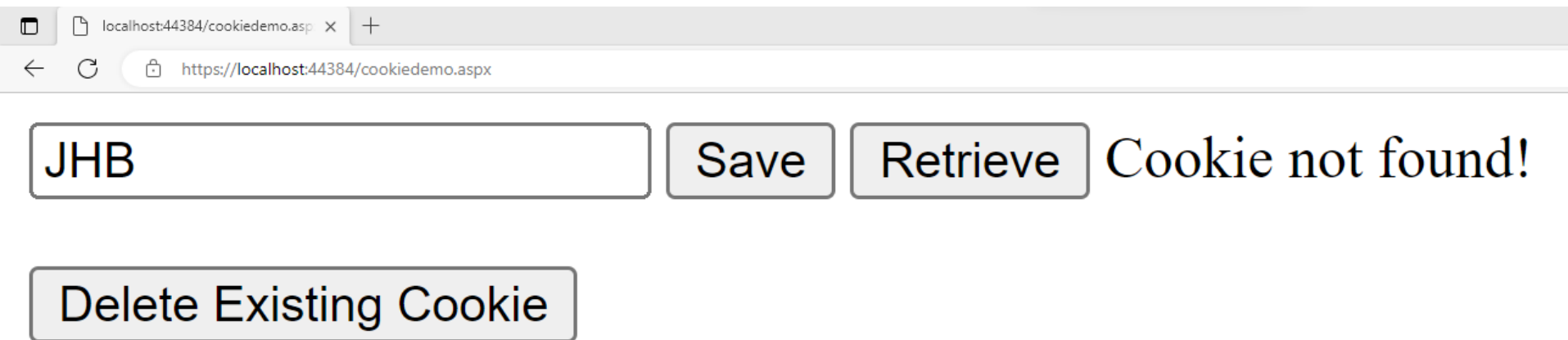
localhost:44384/cookie demo.asp x +

← ↻ https://localhost:44384/cookie demo.aspx

Enter your name Save Retrieve Hello, JHB! Welcome back!

Delete Existing Cookie

Now click Delete and then again click retrieve



The screenshot shows a web browser window with a single tab titled 'localhost:44384/cookie demo.asp'. The address bar displays 'https://localhost:44384/cookie demo.aspx'. The page content includes a text input field containing 'JHB', followed by 'Save' and 'Retrieve' buttons. To the right of these buttons is the text 'Cookie not found!'. Below the input field and buttons is a 'Delete Existing Cookie' button.

localhost:44384/cookie demo.asp × +

← ↻ https://localhost:44384/cookie demo.aspx

JHB Save Retrieve Cookie not found!

Delete Existing Cookie

Implement yourself(session variable demo)



Pizza

French Fries

Sandwich

Burger

Vada Pav

Cold Drink

Submit

When the submit button is clicked, go to final.aspx page, get the details of selected items from the previous page through session variable and display all selected items and final amount.



- Pizza
- Sandwich
- Cold Drink

Final Amount :
475